

GENGP302

APPLIED PHYSICS

Apply General Physics

Competence

RQF Level: 3

Learning Hours



40

Credits: 4

Sector: ALL EXCEPT HOSPITALITY AND TOURISM

Trade: ALL except (Fashion design, Fine and Plastic Arts, Agriculture, Food Processing, Animal Health, Forestry, Leather Technology, Food and Beverages Operations, Front Office and Housekeeping operations, Tourism)

Module Type: General

Curriculum: GENG302-TVET - ALL EXCEPT HOSPITALITY CURRICULA

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Purpose statement	This module describes the knowledge, skills, and attitude required to apply concepts of physics. At the end of this module, the learner will be able to describe basic measurements in physics, analyze motion in one and two dimensions, demonstrate electrostatic phenomena, apply geometric optics and characterize sources of energy in the world.						
	It will help trainee to carry out his/her specialized tasks that are useful in analyzing data, solving real life problems encountered in related fields.						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	● Group project and presentation	20%					
	● Individual project /Work	50%					
				Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe basic measurements in physics	1.1. Physics quantities are clearly derived based on dimension analysis
	1.2. Experimental errors are accurately calculated based on their types
	1.3. Measuring instruments are properly used based on metric system
2. Describe Motion in 1- Dimension	2.1. Displacement, velocity and acceleration concepts are clearly explained in 1 dimension
	2.2. Linear motion concepts are clearly illustrated based on their corresponding graphs
	2.3. Equations of motion are appropriately applied in a straight line under constant acceleration
3. Analyze motion in Two Dimensions	3.1. Scalars, vectors and vector components are clearly described in Cartesian coordinate system
	3.2. Displacement, velocity and acceleration concepts are clearly illustrated in 2 dimensions
	3.3. Motion is clearly described in two dimensions
4. Demonstrate electrostatic phenomena	4.1. Electrostatic charges and their conservation are correctly described in line with laws of static charges on Coulomb's law
	4.2. Electrostatic fields are effectively determined based on Coulomb's law
	4.3. Effects of electric field on charged particles are clearly demonstrated based on fundamental laws of static charges

5. Apply Geometric optics	5.1. Properties of light are properly explained based on homogenous medium
	5.2. Reflection of light on surfaces is effectively applied based on laws of reflection
	5.3. Refraction of light in different media is effectively applied based on Snell's law
6. Characterize sources of energy in the world	6.1. Basic concepts of work, energy and power are properly explained based on the corresponding physical meanings.
	6.2. Types of energy are properly identified based on their sources
	6.3. Relative advantages and disadvantages of various energy sources are properly analyzed according to the sources

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Describe basic measurements in physics 2. Describe Motion in 1- Dimension 3. Analyze motion in Two Dimensions 4. Demonstrate electrostatic phenomena 5. Apply Geometric optics 6. Characterize sources of energy in the world
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Learning outcome 1: Describe basic measurements in physics	Learning hours: 8
Indicative content	
• <i>Derivation of physical quantities</i> ✓ Meaning of fundamental physical Quantities  Mass  Length  Time ✓ Meaning of derived physical quantities  Volume  Weight  Density  Area	



Force

✓ International system of units (SI) and metric prefixes in everyday use



Name, symbol and factor

✓ Dimension analysis



Definition of dimensions of physical quantities



Assigning dimensions to physical quantities



Rules of writing dimensions of physical quantities



Benefits of dimensions



Limitation of dimensions.

● **Calculation of experimental errors**

✓ Types of experimental errors



Definition of experimental error



Systematic error



Random error



Absolute errors



Relative errors



Percentage error

✓ Rules of identifying significant digits



Examples on applying rules of significant figures



Rounding off numbers

● **Using measuring instruments at workplace.**

✓ Measuring instruments used to measure length

✓ Measuring instruments used to measure mass

✓ Measuring instruments used to measure time

Resources required for the learning outcome

Equipment	PPE, Whiteboard and chalkboard
Materials	Chalks, Markers, Flipchart
Tools	Computer, Projector, Various measuring instruments, Textbooks, Scientific calculator
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion

Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment
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Learning outcome 2: Describe Motion in 1- Dimension	Learning hours: 5
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Indicative content	
• Explanation of displacement, velocity and acceleration concepts	
✓ Displacement & distance  Point location on a line  Speed and velocity ✓ Description of Average & Instantaneous acceleration	
• Illustration of linear motion using corresponding graphs	
✓ Slopes and General Relationships  Definition of rise, run, slope, intercept and equation of a straight line ✓ Graph of displacement vs. time  Constant velocity  Constant acceleration  Non constant acceleration ✓ Graph of velocity vs. Time  With positive acceleration  With zero acceleration  With negative acceleration	
• Application of equations of motion	
✓ Velocity as a function of time  Initial velocity  Acceleration  time ✓ Displacement as a function of time  Relation between displacement and average velocity ✓ Final velocity as a function of displacement	

 Equation of velocity  Acceleration ✓ Analyze freely falling objects  Acceleration of Gravity  Representing Free Fall by Graphs	
Resources required for the indicative content	
Equipment	PPE, whiteboard, chalkboard, computer, projector, textbooks
Materials	Chalks, markers
Tools	Scientific calculator, meter ruler, compass
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion • Research
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Learning outcome 3: Analyze motion in two Dimensions	Learning hours: 6
Indicative content	
• Description of scalars, vectors and vector components ✓ Scalars  Properties of scalars  Operation on scalars ✓ Vectors  Properties of vectors	

 Vectors addition & subtraction

✓ Vector components in Cartesian coordinate system

 Point location in space

 position vector

 Components of the position vector.

● Illustration of displacement, velocity and acceleration

✓ Displacements in two dimensions

 Subtraction of vectors

 Change in position vector

✓ Velocity in two dimensions

 Average velocity

 Instantaneous velocity

✓ Acceleration in two dimensions

 Average acceleration

 Instantaneous acceleration

● Description of motion

✓ Motion in two dimensions

 Graphical representation

 Velocity components

✓ Projectile motion

 Definition of projectile motion

 Applications of projectile motion

 Projection angle

✓ Equations of motion

 vertical component

 Horizontal component

Resources required for the indicative content

Equipment	PPE, whiteboard and chalkboard, computer, projector, textbooks
Materials	Chalks, markers
Tools	Scientific calculator, meter ruler, compass
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided

	• Group discussion • Research
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Learning outcome 4: Demonstrate electrostatic phenomena	Learning hours: 8
Indicative content	
• Description of electrostatic charges and their conservation ✓ Electric charges  Elementary charge  Point charge  Sign and magnitude of electric charges  Laws of electrostatic charges  Conservation of electric charges ✓ Electrification (charging) methods  Electrification by rubbing  Electrification by contact  Electrification by induction ✓ Electrostatic field  Electric field lines  Electric field at a point charge  Electric field between two charges • Determination of the electrostatic fields ✓ Coulomb's law of electrostatic charges  Electric permittivity  Coulomb's law  Mathematical treatment of coulombs of electrostatic charges ✓ Electric field intensity and potential  Electric field intensity and its mathematical treatment	

 Electric field strength due to distribution of electric field.

 Electric field potential and its mathematical treatment

✓ Effect of an electrostatic field on a moving charge

 Charge deflection

 Charge acceleration

- **Demonstration of effects of electric field on charged particles**

✓ Capacitors

 Definition of Capacitance

 Parallel-plate capacitor

 Effective capacitance for capacitor network

✓ Electrostatic energy stored by a capacitor

 Description of energy in capacitors

 Calculation of energy stored in capacitors

✓ Examples of electrostatic phenomena

 Electrostatic discharge

 Lightning arrestors

 Paint spraying

 Photocopier machines

Resources required for the indicative content

Equipment	PPE, whiteboard and chalkboard, computer, projector, textbooks
Materials	Chalks, markers
Tools	Scientific calculator, meter ruler, compass
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion • Research
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Indicative content

- **Explanation of light**

- ✓ Key terms

-  Definition of optics
-  Definition of geometrical optics
-  Properties of light

- **Application of light reflection on surfaces**

- ✓ Laws of reflection

-  Definition of light reflection
-  Types of reflection of light
-  Ray diagram for reflection
-  Laws of reflection
-  Parallax method

- ✓ Formation of image in mirrors

-  Plane mirror
-  Spherical mirrors

- ✓ Applications of light reflection

-  Reflecting periscope
-  Car side mirrors

- **Application of light refraction in different media**

- ✓ Key terms

-  Interface
-  Refraction

- ✓ Laws of refraction

-  1st law of refraction
-  Snell's law of refraction

- ✓ Real and apparent depth

- ✓ Total internal reflection

-  critical angle

- ✓ Refraction of light

-  Refraction of light in Prism

Resources required for the indicative content

Equipment	PPE, whiteboard, chalkboard, optical bench, optical slide, computer, projector, textbooks
Materials	Chalks, markers, candles, water
Tools	Scientific calculator, meter ruler, prism, thin lenses, compass
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion • Research
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment • Project based assessment

Indicative content

- **Explanation of basic concepts of work, energy and power**

- ✓ Explanation of work



Positive work



Negative work



Zero work



Net or unbalanced work

- ✓ Explanation of energy



Physical meaning of energy



Mechanical energy



Energy conservation law



Work-energy theorem for linear motion

- ✓ Explanation of power



Meaning of power



Calculation of power



Energy dissipation



Efficiency of machines

- **Identification of types of energy**

- ✓ Energy



Description of energy



source of energy

- ✓ Identification of sources of energy



Renewable sources of energy



Non-renewable sources of energy

- ✓ Formation of energy



Formation of renewable energy systems.



Formation of non-renewable energy systems

- **Analyzing relative advantages and disadvantages of various energy sources**

- ✓ Non- renewable (fossil fuel) energy



Oil

 Coal  Natural gases  Renewable sources  Wind  Solar  Geothermal  Hydropower	
Resources required for the indicative content	
Equipment	PPE, whiteboard and chalkboard, computer, projector, textbooks
Materials	Chalks, markers
Tools	Scientific calculator
Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Trainer guided Group discussion Research
Formative assessment methods	Written assessment Oral presentation Performance assessment

References:

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7. David, h. R. (2011). *Fundamentals of physics 9th edition*. Jefferson: john willey & sons.
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